

CookBook - Your Virtual Kitchen Assistant

Introduction:

Project Title: Cook Book - Your Virtual Kitchen Assistant

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Project Overview

Purpose:

Cook Book is a virtual kitchen assistant designed to simplify cooking by providing users with step-by-step recipes, personalized recommendations, ingredient substitution suggestions, and an organized digital cookbook. It aims to help beginners and food enthusiasts explore, learn, and cook efficiently.

Features:

- User-friendly recipe search and filtering (by cuisine, ingredients, or cooking time).
- Step-by-step cooking instructions with images or videos.
- Personalized recipe recommendations based on preferences.
- Ingredient substitution suggestions.
- Option to save and manage favorite recipes.
- Responsive design for both desktop and mobile devices.

Architecture

Component Structure:

- App.js - Main entry point.
- Navbar - Navigation menu (Home, Recipes, Favorites, About).

- `RecipeList` - Displays a list of recipes.
- `RecipeDetail` - Shows detailed instructions and ingredients.
- `Favorites` - User's saved recipes.
- `SearchBar` - Filters recipes by keywords.

State Management:

- `React Hooks` (`useState`, `useEffect`) for local states.
- `Context API` for managing global favorites and user preferences.

Routing:

- `React Router` for navigation:
 - o `/` → Home Page
 - o `/recipes` → Recipe List
 - o `/recipe/:id` → Recipe Details
 - o `/favorites` → Saved Recipes

Setup Instructions

Prerequisites:

- `Node.js` and `npm` installed.

Installation:

```
git clone <repository-url>
cd cookbook-virtual-assistant
npm install
```

Folder Structure:

- `/src`
 - o `/components` → `Navbar`, `SearchBar`, `RecipeCard`
 - o `/pages` → `Home`, `Recipes`, `Favorites`, `About`
 - o `/assets` → Images, icons, and styles
 - o `/context` → Global context for favorites and preferences

Running the Application

```
npm start
```

Component Documentation

Key Components:

- **RecipeCard** - Displays recipe preview (image, name, short description).
- **RecipeDetail** - Provides full cooking instructions.
- **Favorites** - Stores user-selected recipes.

Reusable Components:

- **Button** - Custom styled buttons.
- **SearchBar** - Search input for recipes.

State Management

- **Global State:** Favorites and user preferences stored using Context API.
- **Local State:** Recipe filters, search queries, and temporary UI states handled using `useState`.

User Interface

- Clean and modern interface.
- Recipe cards with thumbnails.
- Detailed recipe view with stepwise cooking process.

Styling

- Tailwind CSS for fast and responsive design.
- Custom theme with warm food-inspired colors.

Testing

- Unit testing with Jest.
- Component testing with React Testing Library.

Screenshots or Demo

(Add screenshots of the app once developed)

Known Issues

- *Limited dataset of recipes in initial version.*
- *No offline mode in current release.*

Future Enhancements

- *AI-powered personalized meal recommendations.*
- *Voice-assisted cooking guide.*
- *Integration with grocery delivery services.*
- *Offline mode with cached recipes.*